

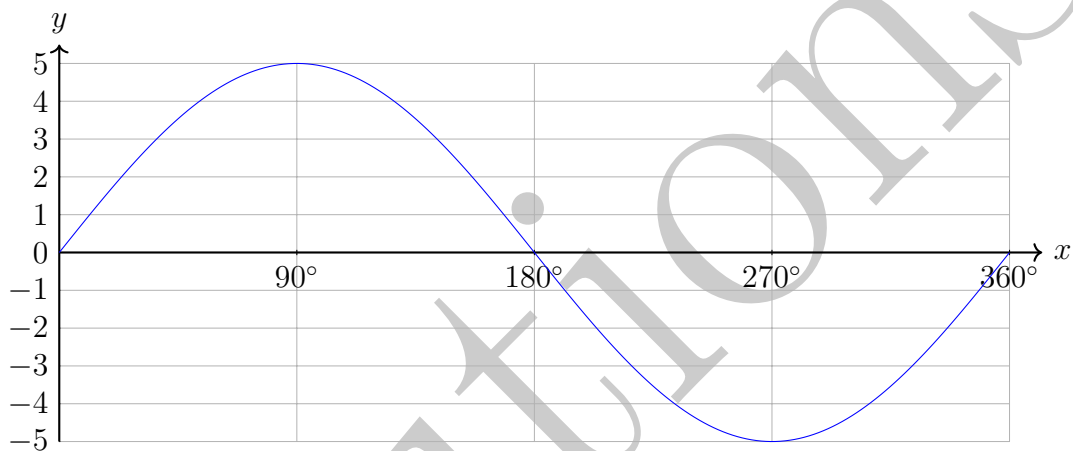
Foundation Mathematics 1017SCG
Week 9 Workshop Solutions

Remember that the method shown in the solutions is not the only way to solve the question. There are usually multiple ways to solve the same question.

1. (a)

$$y = 5 \sin(x)$$

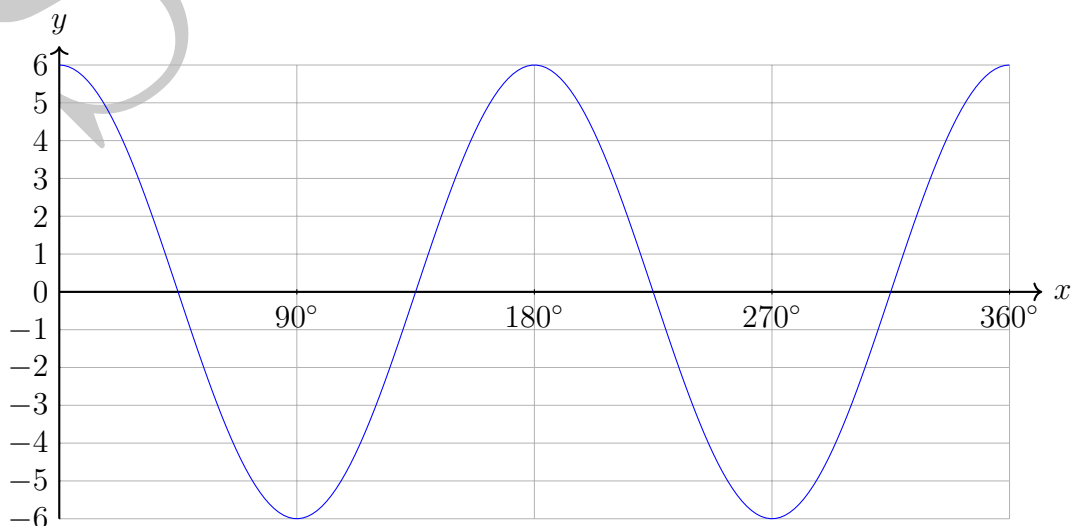
- Amplitude is 5
- Period is 360°
- No horizontal shift
- No vertical shift
- Minimum value of y is -5
- Maximum value of y is 5



(b)

$$y = 6 \cos(2x)$$

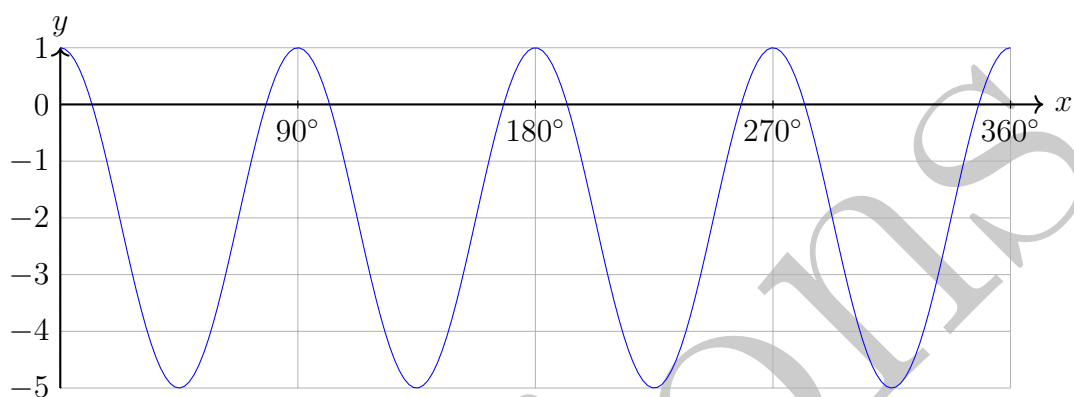
- Amplitude is 6
- Period is 180°
- No horizontal shift
- No vertical shift
- Minimum value of y is -6
- Maximum value of y is 6



(c)

$$y = 3 \cos(4x) - 2$$

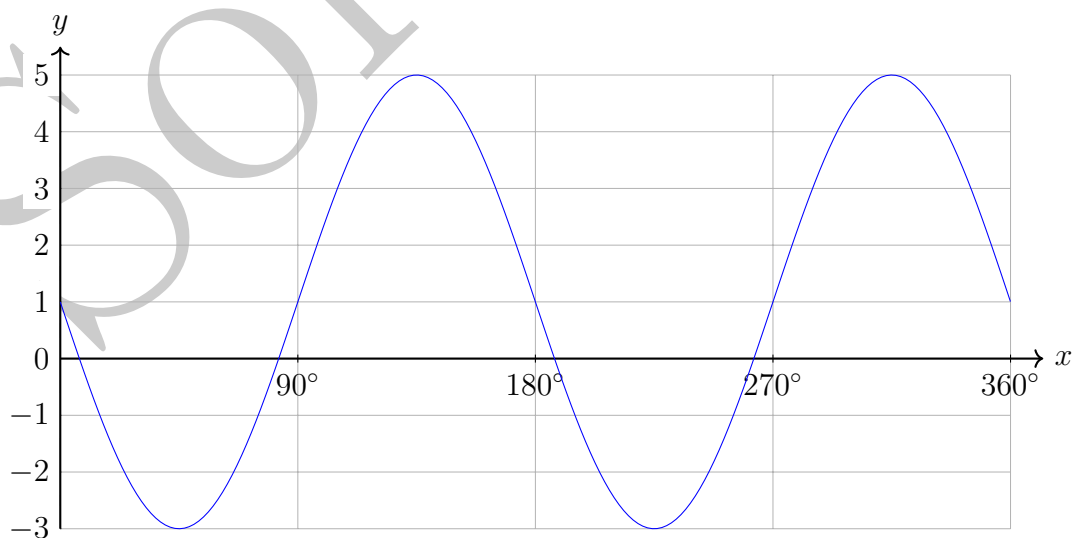
- Amplitude is 3
- Period is 90°
- No horizontal shift
- Vertical shift is 2 down
- Minimum value of y is -5
- Maximum value of y is 1



(d)

$$y = -4 \sin(2x) + 1$$

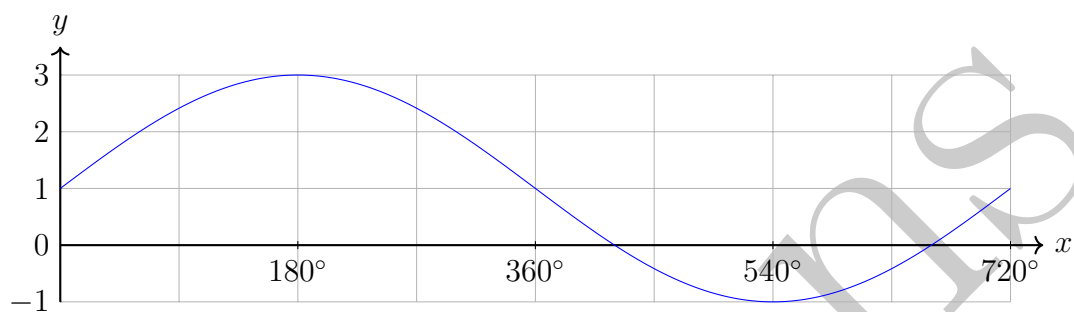
- Note that the graph is reflected across the x -axis
- Amplitude is 4
- Period is 180°
- No horizontal shift
- Vertical shift is 1 up
- Minimum value of y is -3
- Maximum value of y is 5



(e)

$$y = 2 \sin \left(\frac{x}{2} \right) + 1$$

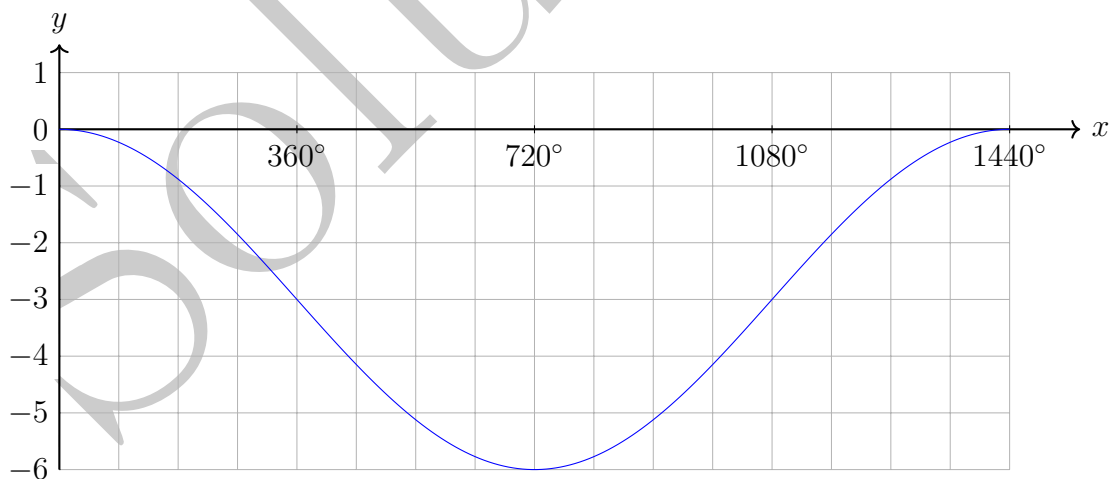
- Amplitude is 2
- Period is 720°
- No horizontal shift
- Vertical shift is 1 up
- Minimum value of y is -1
- Maximum value of y is 3



(f)

$$y = 3 \cos \left(\frac{x}{4} \right) - 3$$

- Amplitude is 3
- Period is 1440°
- No horizontal shift
- Vertical shift 3 down
- Minimum value of y is -6
- Maximum value of y is 0

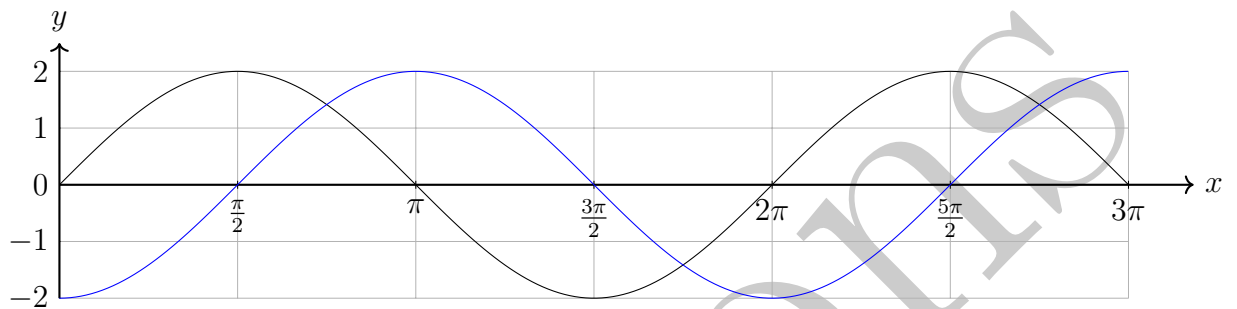


2. (a)

$$y = 2 \sin \left(x - \frac{\pi}{2} \right)$$

- Amplitude is 2
- Period is 2π
- Horizontal shift is $\frac{\pi}{2}$ to the right
- There is no vertical shift

The graph of $y = 2 \sin(x)$ is shown in black and the graph of $y = 2 \sin \left(x - \frac{\pi}{2} \right)$ is shown in blue.

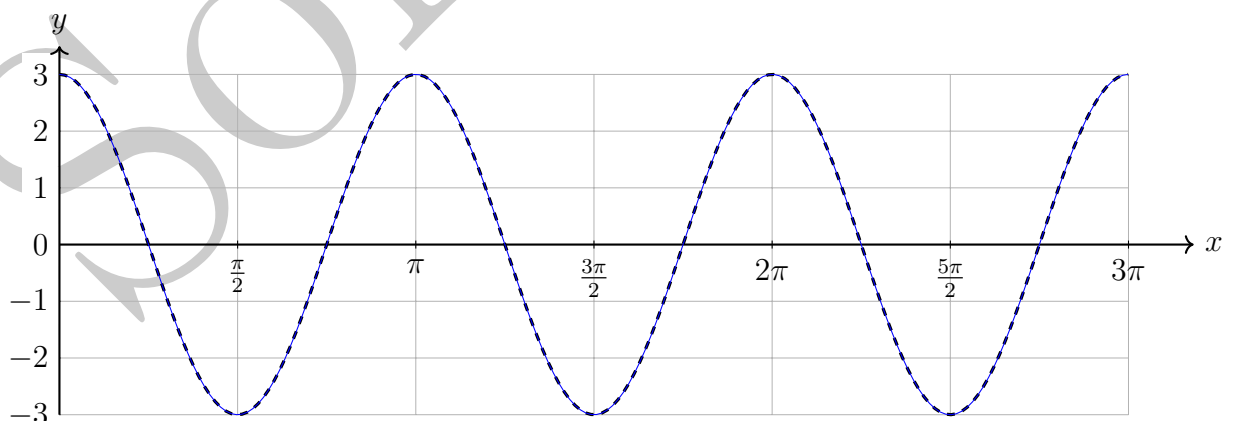


(b)

$$y = 3 \cos[2(x + \pi)]$$

- Amplitude is 3
- Period is π
- Horizontal shift is π to the left
- There is no vertical shift

The graph of $y = 3 \cos(2x)$ is shown in black (dashed line) and the graph of $y = 3 \cos[2(x + \pi)]$ is shown in blue. Note that both graphs look exactly the same. As both graphs have a period of π , shifting the graphing horizontally by π does not change the appearance of the graph.



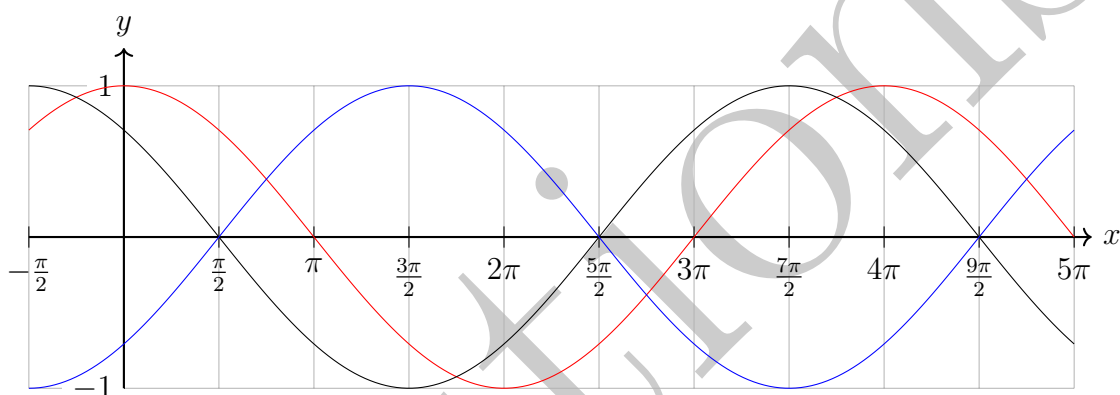
(c)

$$y = -\cos \left[\frac{1}{2} \left(x + \frac{\pi}{2} \right) \right]$$

- Amplitude is 1
- Period is 4π
- Horizontal shift is $\frac{\pi}{2}$ to the left
- There is no vertical shift
- The graph is reflected across the x -axis

As there are several transformations, you may find it helpful to draw several graphs and including one transformation at a time. For example

- Start by drawing a graph of $y = \cos \left(\frac{x}{2} \right)$ - shown in red
- Now draw a graph of $y = \cos \left[\frac{1}{2} \left(x + \frac{\pi}{2} \right) \right]$ - shown in black
- Finally draw a graph of $y = -\cos \left[\frac{1}{2} \left(x + \frac{\pi}{2} \right) \right]$ - shown in blue

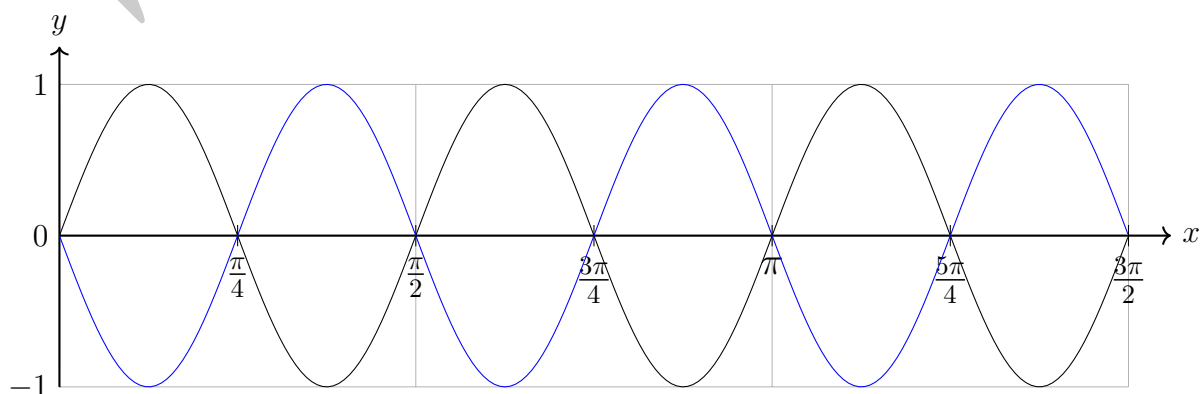


(d)

$$\begin{aligned} y &= \sin(4x - \pi) \\ &= \sin \left[4 \left(x - \frac{\pi}{4} \right) \right] \end{aligned}$$

- Amplitude is 1
- Period is $\frac{\pi}{2}$
- Horizontal shift is $\frac{\pi}{4}$ to the right
- There is no vertical shift

The graph of $y = \sin(4x)$ is shown in black and the graph of $y = \sin \left[4 \left(x - \frac{\pi}{4} \right) \right]$ is shown in blue.

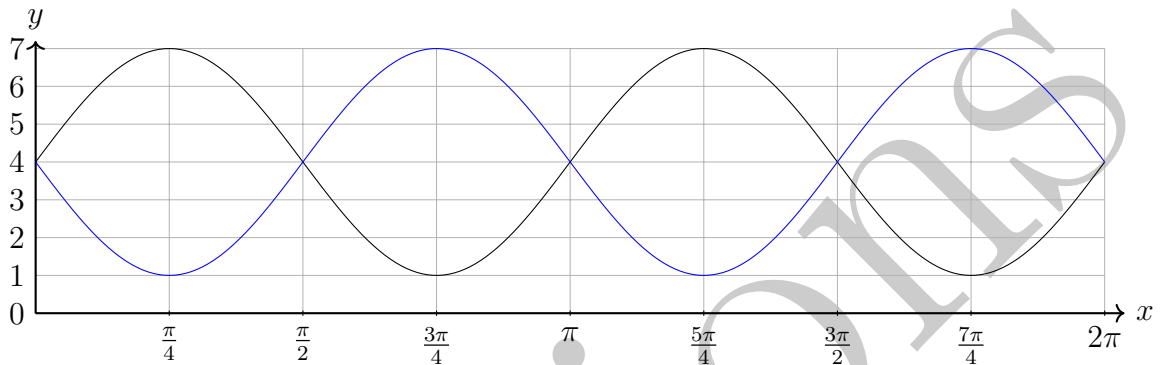


3. (a)

$$y = 3 \sin \left[2 \left(x - \frac{\pi}{2} \right) \right] + 4$$

- Amplitude is 3
- Period is π
- Horizontal shift is $\frac{\pi}{2}$ to the right
- Vertical shift is 4 up

The graph of $y = 3 \sin(2x) + 4$ is shown in black and the graph of $y = 3 \sin \left[2 \left(x + \frac{\pi}{2} \right) \right] + 4$ is shown in blue.

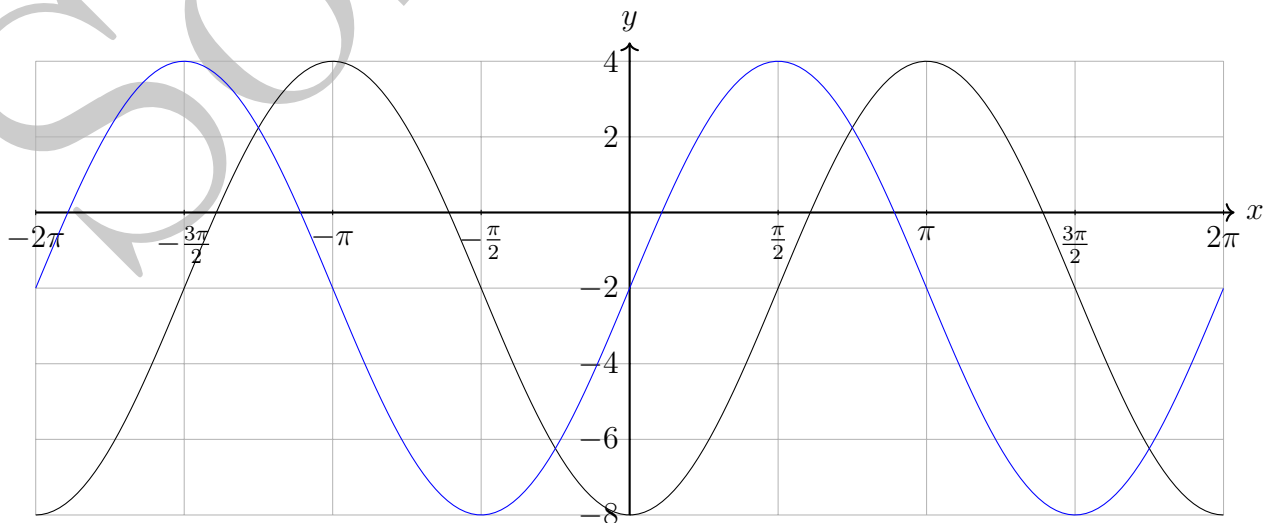


(b)

$$y = -6 \cos \left(x + \frac{\pi}{2} \right) - 2$$

- Amplitude is 6
- Period is 2π
- Horizontal shift is $\frac{\pi}{2}$ to the left
- Vertical shift is 2 down
- The graph is reflected across the x -axis

The graph of $y = -6 \cos(x) - 2$ is shown in black and the graph of $y = -6 \cos \left(x + \frac{\pi}{2} \right) - 2$ is shown in blue.



4. (a) The amplitude is 2 and the vertical shift is 1 up. Therefore $A = 2$ and $D = 1$. The period of the graph is 2π which makes $B = 1$.

$$y = 2 \cos(x) + 1$$

- (b) The amplitude is 3 and the vertical shift is 1 down. Therefore $A = 3$ and $D = -1$. The period of the graph is π which makes $B = 2$.

$$y = 3 \sin(2x) - 1$$

- (c) The amplitude is 3 and the vertical shift is 2 up. Note that the graph starts at its minimum value. This means that it is a cos graph, but has been reflected across the x -axis. Therefore $A = -3$ and $D = 2$. The period of the graph is $\frac{\pi}{2}$ which makes $B = 4$.

$$y = -3 \cos(4x) + 2$$

- (d) The amplitude of the graph is 5 and the vertical shift is 2 up. Therefore $|A| = 5$ and $D = 2$. The period of the graph is π which makes $B = 2$. Below are a few examples of equations that satisfy the graph, but there are many more.

Writing the graph as a cos graph:

$$y = 5 \cos(2x) + 2$$

$$y = 5 \cos[2(x + \pi)] + 2$$

$$y = 5 \cos[2(x - \pi)] + 2$$

$$y = -5 \cos\left[2\left(x - \frac{\pi}{2}\right)\right] + 2$$

$$y = -5 \cos\left[2\left(x + \frac{\pi}{2}\right)\right] + 2$$

Writing the graph as a sin graph:

$$y = 5 \sin\left[2\left(x - \frac{3\pi}{4}\right)\right] + 2$$

$$y = 5 \sin\left[2\left(x + \frac{\pi}{4}\right)\right] + 2$$

$$y = -5 \sin\left[2\left(x - \frac{\pi}{4}\right)\right] + 2$$

$$y = -5 \sin\left[2\left(x + \frac{3\pi}{4}\right)\right] + 2$$

5. (a)

$$\begin{aligned} \lim_{x \rightarrow 3} x^2 + 1 &= 3^2 + 1 \\ &= 10 \end{aligned}$$

- (b)

$$\begin{aligned} \lim_{x \rightarrow 4} 2x - 8 &= 2 \times 4 - 8 \\ &= 0 \end{aligned}$$

(c)

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{2x+1}{x} &= \frac{2 \times 2 + 1}{2} \\ &= \frac{5}{2}\end{aligned}$$

(d)

$$\begin{aligned}\lim_{x \rightarrow -1} \frac{3x^2+1}{x-1} &= \frac{3 \times (-1)^2 + 1}{-1 - 1} \\ &= \frac{4}{-2} \\ &= -2\end{aligned}$$

(e)

$$\begin{aligned}\lim_{x \rightarrow 5} \frac{x^2 - 5x}{x - 5} &= \lim_{x \rightarrow 5} \frac{x(x-5)}{(x-5)} \\ &= \lim_{x \rightarrow 5} x \\ &= 5\end{aligned}$$

(f)

$$\begin{aligned}\lim_{x \rightarrow -3} \frac{x^2 + 5x + 6}{x + 3} &= \lim_{x \rightarrow -3} \frac{(x+2)(x+3)}{x+3} \\ &= \lim_{x \rightarrow -3} x + 2 \\ &= -3 + 2 \\ &= -1\end{aligned}$$

(g)

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{7x - x^2}{x} &= \lim_{x \rightarrow 0} \frac{x(7-x)}{x} \\ &= \lim_{x \rightarrow 0} 7 - x \\ &= 7 - 0 \\ &= 7\end{aligned}$$

(h)

$$\begin{aligned}\lim_{x \rightarrow -3} \frac{5x+15}{x^2-9} &= \lim_{x \rightarrow -3} \frac{5(x+3)}{(x+3)(x-3)} \\ &= \lim_{x \rightarrow -3} \frac{5}{x-3} \\ &= \frac{5}{-3-3} \\ &= -\frac{5}{6}\end{aligned}$$

(i)

$$\begin{aligned}\lim_{x \rightarrow 4} \frac{2x^3 - 8x^2}{x - 4} &= \lim_{x \rightarrow 4} \frac{2x^2(x - 4)}{x - 4} \\ &= \lim_{x \rightarrow 4} 2x^2 \\ &= 2 \times 4^2 \\ &= 32\end{aligned}$$

(j)

$$\begin{aligned}\lim_{x \rightarrow -1} \frac{x^2 + 6x + 5}{x^3 + x^2} &= \lim_{x \rightarrow -1} \frac{(x + 1)(x + 5)}{x^2(x + 1)} \\ &= \lim_{x \rightarrow -1} \frac{x + 5}{x^2} \\ &= \frac{-1 + 5}{(-1)^2} \\ &= \frac{4}{1} \\ &= 4\end{aligned}$$

(k)

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{x^2 + 4x - 12}{x^2 - 3x + 2} &= \lim_{x \rightarrow 2} \frac{(x - 2)(x + 6)}{(x - 1)(x - 2)} \\ &= \lim_{x \rightarrow 2} \frac{x + 6}{x - 1} \\ &= \frac{2 + 6}{2 - 1} \\ &= \frac{8}{1} \\ &= 8\end{aligned}$$

6. (a)

$$f(x) = \frac{x^2 - 25}{x - 5}$$

The function $f(x)$ is not continuous at $x = 5$ as $f(5)$ is undefined.

(b)

$$f(x) = \begin{cases} 4 - x & -\infty < x \leq -1 \\ 3x + 1 & -1 < x < \infty \end{cases}$$

The piecewise function consists of two linear pieces. As linear functions are continuous, the only place the function may be discontinuous is at $x = -1$. To determine if the function $f(x)$ is continuous at $x = -1$, we need to test the three criteria.

As $f(-1) = 4 - (-1) = 5$, the first criteria is satisfied.

$$\begin{aligned}
\lim_{x \rightarrow -1^-} f(x) &= \lim_{x \rightarrow -1^-} 4 - x \\
&= 4 - -1 \\
&= 5 \\
\lim_{x \rightarrow -1^+} f(x) &= \lim_{x \rightarrow -1^+} 3x + 1 \\
&= 3 \times -1 + 1 \\
&= -2
\end{aligned}$$

As $\lim_{x \rightarrow -1^-} f(x) \neq \lim_{x \rightarrow -1^+} f(x)$, then $\lim_{x \rightarrow -1} f(x)$ does not exist. Therefore $f(x)$ is not continuous at $x = -1$.

(c)

$$f(x) = \begin{cases} 4 - x^2 & -\infty < x < -2 \\ x + 2 & -2 \leq x < \infty \end{cases}$$

The piecewise function is made consists of a linear piece and a quadratic piece. As linear functions and quadratic functions are continuous, the only place where function may be discontinuous is at $x = -2$.

As $f(-2) = -2 + 2 = 0$, the first criteria is satisfied.

$$\begin{aligned}
\lim_{x \rightarrow -2^-} f(x) &= \lim_{x \rightarrow -2^-} 4 - x^2 \\
&= 4 - (-2)^2 \\
&= 0 \\
\lim_{x \rightarrow -2^+} f(x) &= \lim_{x \rightarrow -2^+} x + 2 \\
&= -2 + 2 \\
&= 0
\end{aligned}$$

Therefore, we have

$$\lim_{x \rightarrow -2} f(x) = 0 = f(-2)$$

Therefore, criteria two and three are also satisfied and the function $f(x)$ is continuous.